Before the changes, Messflix was struggling with tight coupling between systems owned by system of records teams, the Data team, and teams developing dashboards. Any change introduced to any system was problematic, because tight connections between systems created an overhead cost to adapt all upstream systems. Systems had to evolve together, slowing them all to a standstill. In addition, the Data team workload created a bottleneck affecting the whole organization.

As you can see in figure 4.18, you divide the responsibilities of the Data team among teams that were closest to the data. You enable the reuse of data—for example, the Movie Trends data product used by marketing is reused as an input for scripts recommendation by ML. You release data from silos. The Cast, Movie Popularity, Scripts, and Movie Trends data products enable the creation of the script recommender, enabling new business opportunities. You change chaotically stored production cost statement files into a data product. That simplifies the integration with other systems, such as the ERP and Financial Analysis data product (formerly the data warehouse). You analyze use cases and notice that customer 360 data unnecessarily squeezes into a Financials data product. There is no actual use for customer 360 data, so you can simply remove it and leave the Financials data product with a single purpose.

These changes simplify integration from a technical perspective and improve team collaboration from a social perspective. The changes make the overall architecture a well-working, optimized, and decentralized socio-technical architecture.

4.5.2 Data products form a mesh

As you can see in figure 4.19, data products have input and output ports. In addition, some of the output ports can be consumed by one or more data products or other systems. Therefore, data products derived from other data products are a perfectly viable solution.

Similar to microservices, we can decompose the landscape into smaller pieces: data products. If we orient them around their small subdomains (figure 4.19) and connect their output and input ports, we see that data products form a mesh—a data mesh?